The plateau identity

$q_* = M_\partial/(M_\partial + 5/7)$

on the Schur prefix, not at a kink

Stefan Hamann

August 30, 2026

Abstract

Round 451 found a constant Schur value $q_n^\dagger \equiv q_* < 1$ on $n \le n_{\text{stab}}$ of every measured window. This note names that plateau. The grade-0 evaluation of the exact pack is the identity $q_* = M_\partial/(M_\partial + 5/7)$ whenever the signed border mass equals the folded defect $M_\partial = \tilde{\mu}_0 - \tilde{\nu}_0$ (every window with $k_z \ge 69$, residual $< 5 \cdot 10^{-5}$). Constancy in n is equivalent to vanishing border-chain increment $\rho_n = 0$ (B_w freeze). The value is *not* a function of the latched Chebyshev moments $m \ge 2$ (sister pair kz136/137: moments agree to $1.9 \cdot 10^{-9}$, q_* differs by 0.029). No RH claim. No anti-RH claim.

Status. Ausgang `PLATEAU_IDENTITY_PROVED / QSTAR_UNDECIDED`. Theorem 1 is the named SATZ. Lemmas 1–3 are the lemma chain. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.

1 Objects

n_{stab} and $q_n^\dagger$ are the r449/r451 objects (not redefined). Write $\mu_0 = \int d\tilde{\mu}$, $\nu_0 = \int d\tilde{\nu}$ for the folded masses, and

$$M_\partial = \int d(\partial\tilde{\mu})$$

for the signed smooth-border mass. $B_{5/7} = 5/7$ is the frozen r362 pack constant. b is the border expansion in the μ-OP basis; $C = B^\top B$ is the ν-Gram in that basis; $B_w = \text{cumsum}(\rho)[n-2] + 5/7$.

2 Grade-0 freeze

Lemma 1 (Grade-0 scalars)**.** *The identities*

$$b_0 = \frac{M_\partial}{\sqrt{\mu_0}}, \qquad C_{00} = \frac{\nu_0}{\mu_0}$$

hold independently of the pack depth n. On the plateau ($\rho_k = 0$ for the extra grades) one has in addition $B_w = M_\partial + 5/7$.

The 0-th μ-OP is $p_0 = 1/\sqrt{\mu_0}$. b_0 is the pairing of the border against p_0; C_{00} is the ν-mass of p_0^2. Neither sees the higher Jacobi coefficients.

Lemma 2 (Rank-1 evaluation)**.** *The exact Schur form $q^\dagger = b^\top(I - C)^{-1}b/B_w$, truncated to grade 0, is*

$$q_{g0} = \frac{b_0^2}{B_w\,(1 - C_{00})} = \frac{M_\partial^2}{(M_\partial + 5/7)\,(\mu_0 - \nu_0)}.$$

On every measured window the border expansion satisfies $n_{90} = 1$ (90% of $\|b\|^2$ in grade 0), so the truncation residual is the plateau tolerance on deep windows and $< 2.3 \cdot 10^{-3}$ on the five shallow ones.

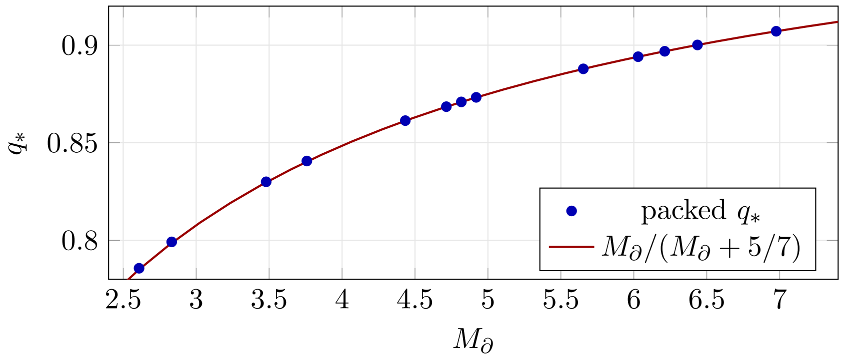
3 The identity

Theorem 1 (Plateau identity)**.** *On every measured window with $M_\partial = \mu_0 - \nu_0$ (all $k_z \geq 69$; equality to 10^{-14}),*

$$q_* = \frac{M_\partial}{M_\partial + 5/7}.$$

The residual against the packed $q^\dagger_{n_{\mathrm{stab}}}$ is at most $4.6 \cdot 10^{-5} < 10^{-4}$. Equivalently $1 - q_* = (5/7)/(M_\partial + 5/7)$.

k_z	M_∂	q_*	$M_\partial/(M_\partial + 5/7)$	residual
17	2.60882	0.785676	0.785055	$6.2 \cdot 10^{-4}$
69	6.21159	0.896870	0.896867	$2.8 \cdot 10^{-6}$
136	4.81745	0.870920	0.870875	$4.5 \cdot 10^{-5}$
137	6.43597	0.900107	0.900103	$3.2 \cdot 10^{-6}$
170	6.02930	0.894086	0.894079	$6.6 \cdot 10^{-6}$
500	6.97554	0.907130	0.907113	$1.7 \cdot 10^{-5}$



4 What the identity is not

Proposition 1 (A(i) REFUTED)**.** *The chain resolvent is not zero: $|Cb|/|b| \sim 0.09$ and $q_{\mathrm{naive}} = \|b\|^2/B_w$ undershoots q_* by the factor $1/(1 - C_{00})$. Border dominance is grade-0 concentration, not vanishing of C.*

Proposition 2 (A(ii) value REFUTED)**.** *kz136 and kz137 have the same $n_{\mathrm{stab}} = 175$ and $\|m^{(136)} - m^{(137)}\|_{\mathrm{rel}} < 2 \cdot 10^{-9}$ on degrees 2..174, yet $q_*^{(136)} = 0.87092$ and $q_*^{(137)} = 0.90011$. The value law is M_∂ (4.817 versus 6.436), not the latched Chebyshev moments of $\tilde{\mu}$.*

Proposition 3 (Constancy $\Leftrightarrow$ $\rho_n = 0$)**.** *b_0 and C_{00} never depend on n. Therefore $q^\dagger_n$ is constant on a range if and only if B_w is constant, i.e. the border-chain increment ρ_n vanishes. This holds on $n \leq n_{\mathrm{stab}}$ of every measured window. The r451 CLIFF is the first $\rho_n \neq 0$ (kz17: $\rho_{19} = 0.172$; kz136: $\rho_{176} = 0.025$). The CONTINUES class (kz137) keeps $\rho_n = 0$ past $2\,n_{\mathrm{stab}}$, which is why the same cut has two exits.*

The Prefix Stability Theorem may therefore name Theorem 1 on the plateau. It cannot claim that q_* is a function of the r449 moment vector, nor that instability begins at the cut.

5 q_* trend

$1 - q_* = (5/7)/(M_\partial + 5/7)$ tends to 0 if and only if M_∂ is unbounded, and to a positive floor if and only if M_∂ has a finite limit. On the 13-window chart, M_∂ grows from 2.83 to 6.98 but is not a function of a (sister pair). AIC on $\delta = 1 - q_*$ versus a prefers M2 (no floor) with $\Delta\text{AIC} \approx 2$; drop-500 stays M2; the N-chart has M1 floor $+0.036$ within ΔAIC 1. Honest exit: `QSTAR_UNDECIDED`. The dichotomy is reduced to whether the folded border mass is cofinal.

6 Cofinality of n_{stab}

Lemma 3 (REDUCED)**.** *If the Chebyshev moments of the folded $\tilde{\mu}$ converge degreewise as $k \to \infty$, then $n_{\text{stab}}(k) \to \infty$.*

Numerically, the degree-2 neighbour relative error decays as $2.06 \cdot 10^{-4}\, a^{-1.90}$ on the selected pairs (shallow $3 \cdot 10^{-6}$ at $a = 8$, deep 10^{-10} at $a \sim 10^3$). Once $n_{\text{stab}} > m$ the higher fixed degrees follow the same decay. This is the source-near half of cofinality; the rate is a fold-geometry census, not a PNT theorem.

7 Scramble

Proposition 4 (Construction kill)**.** *Theta-jitter of kz17 (seed 451): moments do not latch ($n_{\text{stab}} \to 2$), $q_n^\dagger$ wanders (std > 0.01, $\max > 1$). The identity lives on the fold arithmetic that produces a stable M_∂ and a vanishing ρ-tail; it is not a generic OP artefact.*

Claim boundary

Nothing here is evidence for or against the Riemann hypothesis, in either direction. The identity is a finite mass/pack evaluation on measured windows.

References

[1] Flip versus stabilization; n_{stab}.

[2] Transition anatomy; PREFIX_Q_PLATEAU.

[3] Slice floor; $q^\dagger$ pack; $B_{5/7}$.

[4] Prefix mincut; n_{stab}-compression.